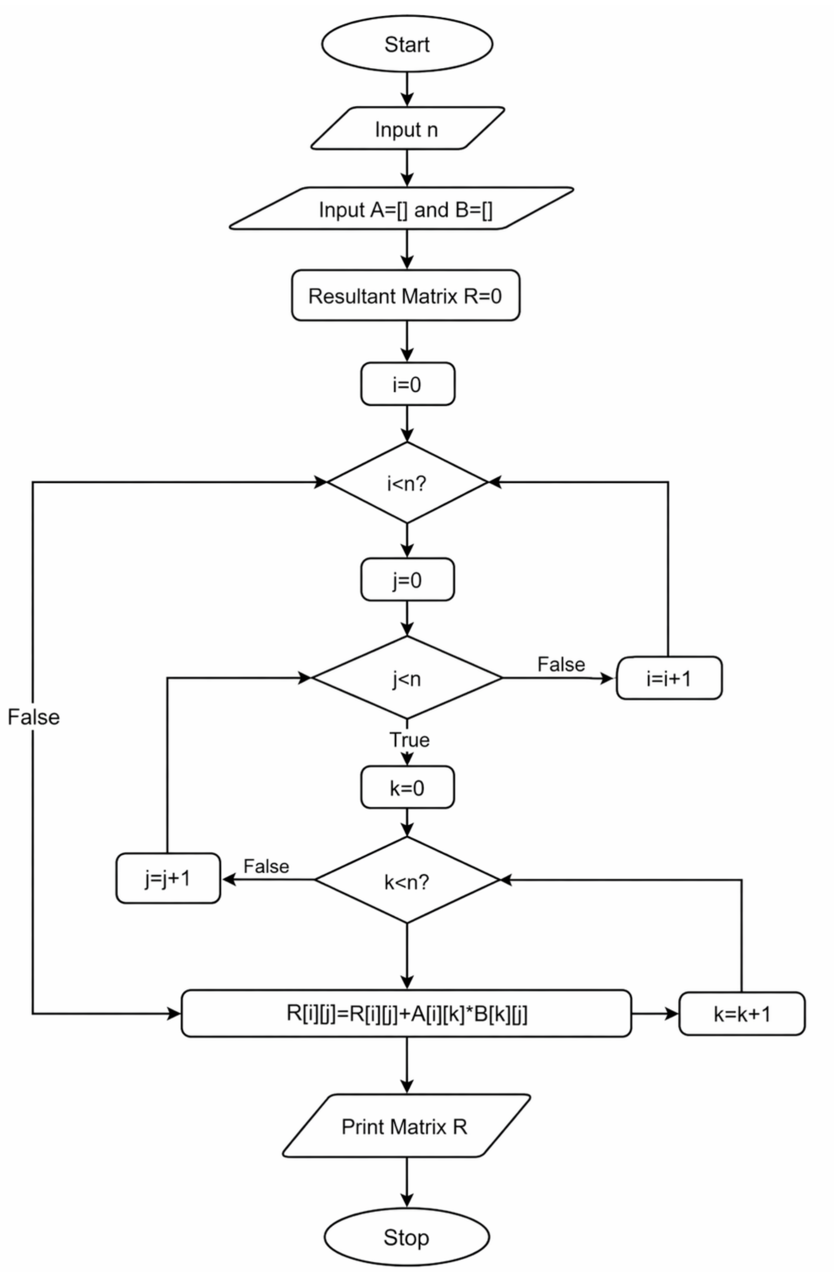
12.1.1. Phone Number Database Management

Aim:Write a Python program to manage a phone number database using a dictionary. The dictionary stores contact names as keys and phone numbers as values.

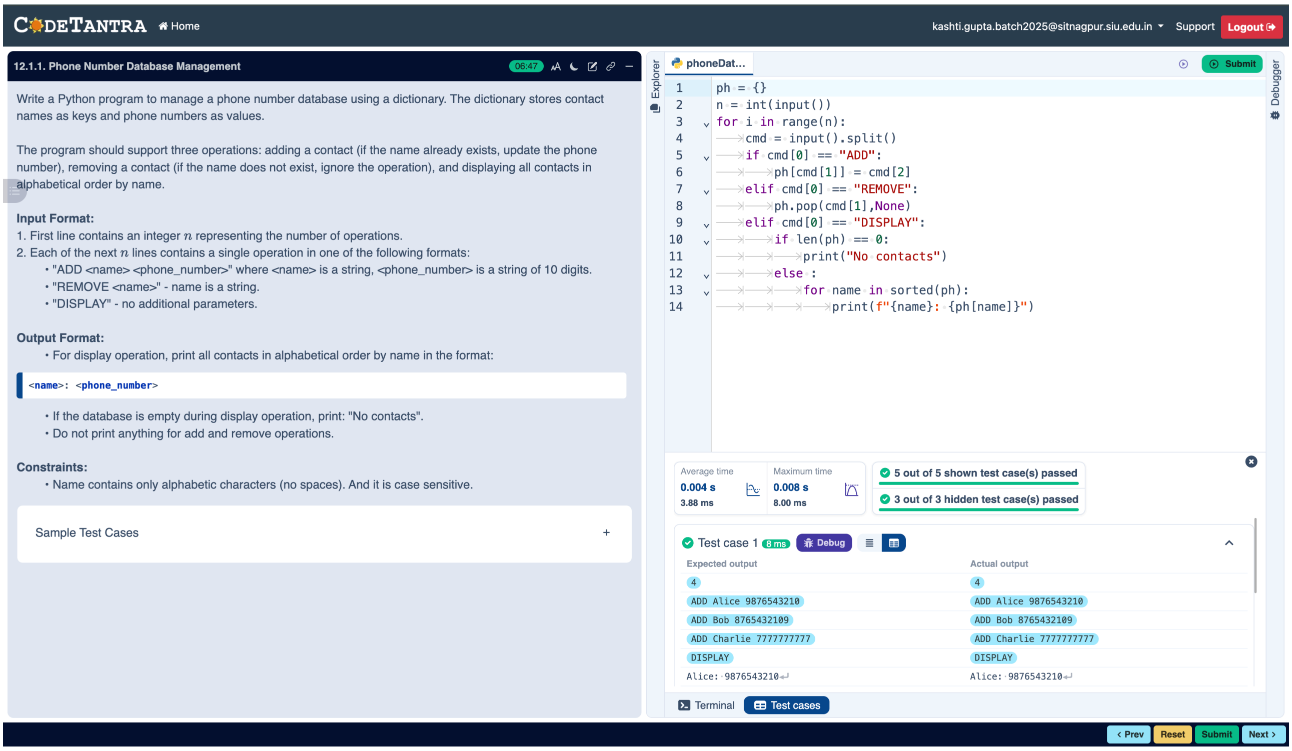
Algorithm

1.Start
2.Create empty dictionary
3.Read number of operations
4.For each operation:
If ADD → insert contact
If REMOVE → delete contact
If DISPLAY → print all contacts (sorted) or "No contacts"
5.Stop

FLOWCHART



EXECUTION


CODETANTRA
Home
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Support
Logout
12.1.1. Phone Number Database Management
06:47
Write a Python program to manage a phone number database using a dictionary. The dictionary stores contact names as keys and phone numbers as values.
The program should support three operations: adding a contact (if the name already exists, update the phone number), removing a contact (if the name does not exist, ignore the operation), and displaying all contacts in alphabetical order by name.
Input Format:
1. First line contains an integer n representing the number of operations.
2. Each of the next n lines contains a single operation in one of the following formats:
• "ADD <name> <phone_number>" where <name> is a string, <phone_number> is a string of 10 digits.
• "REMOVE <name>" - name is a string.
• "DISPLAY" - no additional parameters.
Output Format:
• For display operation, print all contacts in alphabetical order by name in the format:
<name>: <phone_number>
• If the database is empty during display operation, print: "No contacts".
• Do not print anything for add and remove operations.
Constraints:
• Name contains only alphabetic characters (no spaces). And it is case sensitive.
Sample Test Cases
phoneDat...
Submit
ph = {}
n = int(input())
for i in range(n):
    cmd = input().split()
    if cmd[0] == "ADD":
        ph[cmd[1]] = cmd[2]
    elif cmd[0] == "REMOVE":
        ph.pop(cmd[1],None)
    elif cmd[0] == "DISPLAY":
        if len(ph) == 0:
            print("No contacts")
        else :
            for name in sorted(ph):
                print(f"{name}: {ph[name]}")
Average time
0.004 s
3.88 ms
Maximum time
0.008 s
8.00 ms
5 out of 5 shown test case(s) passed
3 out of 3 hidden test case(s) passed
Test case 1
8 ms
Debug
Expected output
4
ADD Alice 9876543210
ADD Bob 8765432109
ADD Charlie 7777777777
DISPLAY
Alice: 9876543210
Actual output
4
ADD Alice 9876543210
ADD Bob 8765432109
ADD Charlie 7777777777
DISPLAY
Alice: 9876543210
Terminal
Test cases
Prev
Reset
Submit
Next